

SECTION A

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		(i) Correct answer only = 14 p[F] (1) (ii) Correct equation used i.e. $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ (1) Correct answer = 1.14 p[F] (1) (iii) Correct use of $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$ for 2 pF and 4 pF (1) Adding 8 (pF) (1) Final answer = 9.33 p[F] (1) (allow 1 mark for 3.43 pF or $\frac{24}{7}$) Deduct 1 mark max if answers expressed as fractions	1	1		6	5	
	(b)		Area of 0.42×0.42 seen [0.1764] (1) Rearrangement i.e. $d = \frac{\epsilon_0 A}{C}$ (1) Separation = 5.78×10^{-7} [m] OR $C = 15.6$ [nF] ish (coming from using $d = 1$ mm) (1) Unrealistic - too thin OR capacitance too small - not achievable (1) ecf			4	4	3	
	(c)		Increases	1			1		
			Question 1 total	4	3	4	11	8	0